Microwave Technology

The Solutions for E3.18 and AO12, 2016

Model answer to Q 1(a): Bookwork

i) Faraday Law of Electromagnetic Induction

$$\nabla x\hat{E} = -\frac{\partial \hat{B}}{\partial t} \Rightarrow -j\omega\mu\hat{H} \quad [V/m]$$

where constitutive relationship, $\hat{B}\,[T \text{ or } Wb/m^2] = \mu\,[H/m]\,\hat{H}\,[A/m]$

i.e. electromotive force induced in a closed circuit is proportional to the rate of change in the magnetic flux density threading the circuit.

[3]

ii) Ampere's Law of Magnetomotive Force

$$\nabla x\hat{H} = \hat{J}c + \hat{J}_D \Rightarrow \sigma_o\hat{E} + \hat{J}_D \quad [A/m^2]$$

where $\hat{J}c\,[A/m^2] = \sigma_o\,[S/m]\,\hat{E}\,[V/m] \Rightarrow 0$ with in[illegible]fect in[illegible]lator

i.e. conduction current creates a closed loop of magne[illegible]c fie[illegible] Note that [illegible]nduction current is simply a surface current density, where the surface [illegible]rans[illegible] to the d[illegible]ection of current flow (i.e. *width x depth*). In addition, Maxwell discovered tha[illegible]a ma[illegible]tic fie[illegible]can also be created by a displacement current. Note that displacemen[illegible] elec[illegible]c flux d[illegible]y) is simply a surface charge density.

$$\hat{J}_D = \frac{\partial \hat{D}}{\partial t} = \varepsilon\frac{\partial \hat{E}}{\partial t} \Rightarrow j\omega\varepsilon\hat{E} \quad [A/[illegible]]$$

where constitutive relati[illegible]hip, $D\,[C/m[illegible]] = \varepsilon\,[F/m]\,\hat{E}\,[V/m]$

For example, when a [illegible]e-varying [illegible]nduc[illegible] current flows through the leads of a parallel-plate capacito[illegible] an equal d[illegible]lacement "wireless" current must also flow between its plates, thus creating a [illegible]se[illegible]p of mag[illegible]tic field between the plates.

[3]

iii) Gauss's Law: [illegible]c form

$$\nabla\cdot\hat{D} = \rho \Rightarrow 0 \quad \text{w[illegible] no stored charges} \quad [C/m^3]$$

i.e. an E-field is created by a stored electric charge. Note that a volume charge density is used here.

[2]

iv) Gauss's Law: Magnetic form

$$\nabla\cdot\hat{B} = 0 \quad [Wb/m^3]$$

i.e. a H-field is not created by a stored magnetic charge. Therefore, it must exist in a closed loop.

[2]

Answers

Model answer to Q 1(b): Bookwork

The electromagnetic wave is periodic in both time (t) and space (z), i.e. $e^{(j\omega t-\gamma z)}$ where $\gamma = jq$ and q = complex wavenumber.

Therefore:

$$\nabla x\hat{E} = -\frac{\partial \hat{B}}{\partial t} \Rightarrow -jqE = -j\omega\mu H$$

$$\therefore \eta = \frac{E}{H} = \frac{j\omega\mu}{\gamma} = \frac{j\omega\mu}{jq}$$

$$but, q = \frac{2\pi}{\lambda} = \frac{\omega\sqrt{\varepsilon_r}}{c} = \omega\sqrt{\mu\varepsilon_o\varepsilon_r}$$

$$\therefore \eta = \sqrt{\frac{\mu_o\mu_r}{\varepsilon_o\varepsilon_r}}$$

[3]

Model answer to Q 1(c): Bookwork

$$\gamma = \frac{j\omega\mu}{\eta} = \alpha + j\beta$$

$$skin\ depth = \frac{1}{\alpha}$$

$$wavelength = \frac{2\pi}{\beta}$$

[2]

Model answer to Q 1(c): Bookwork

Given ε_r = 53 and σ_o = [illegible] at 15 GHz

It is assumed that the distilled water is treated as a homogeneous and isotropic material. Also, distilled water is a non-magnetic material and the following first order approximation is assumed.

$$\varepsilon_r'' = \frac{\sigma_o}{\omega\varepsilon_o}.$$

$$\varepsilon_o = \frac{1}{c^2\mu_o} \quad and\ c = 3\times10^8\ m/s \quad and\ \mu_o = 4\pi\times10^{-7}\ H/m$$

$$\therefore \varepsilon_o = 8.842\ pF/m$$

$$\therefore \varepsilon_r'' = 37.2$$

$$\therefore \eta = \frac{377}{\sqrt{53 - j37.2}} = 44.7 + j14.1\ \Omega$$

$$\therefore \gamma = \frac{j\omega\mu}{\eta} = 762 + j2411 \quad \therefore \delta = 1.31\ mm \quad and \quad \lambda = 2.61\ mm$$

[5]

Answers

Model answer to Q 2(a): Bookwork Derivation

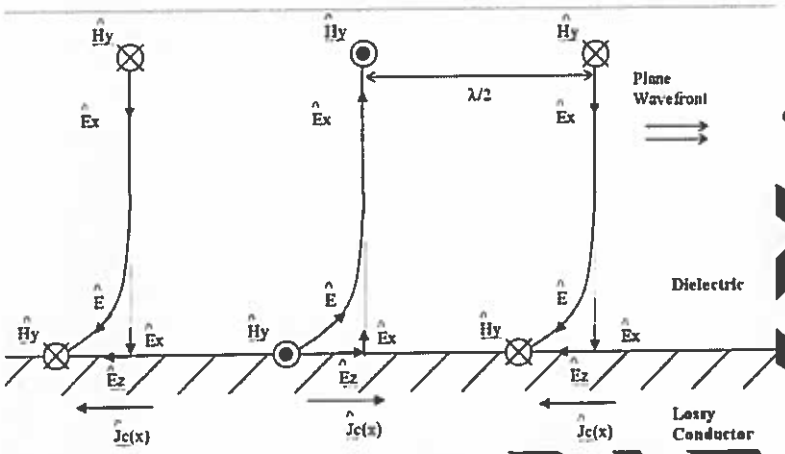
Ez(x=0) = Zs Js, where Js = n x Hy and n = normal unit vector and Zs = surface impedance

Therefore, when σo is made finite, a tangential electric field exists, since the surface impedance is no longer zero.

[2]

Model answer to Q 2(b): Bookwork

Since Ez(0) exists, the resultant E-field in the normal metal leans forward, just above the surface of the conductor, i.e. E = xEx + zEz. Therefore, just above the surface, the wave is not pure TEM, because the E-field, H-field and direction of propagation are not mutually orthogonal.



[5]

Model answer to Q 2(c): Bookwork Derivation

Well above the surface the Poynting vector Pz = Ex x Hy.
Now, since Ez and Hy exist inside the conductor, a wave can propagate inside this material, i.e. with Poynting vector Px(x) = Ez(x) x Hy(x), where Ez(x) = Ez(0)e-γx and Hy = Ez/Zs. If a wave propagates inside the material, the associated E-field will induce a conduction current,
Jc(x) = σo Ez(x) = σo Ez(0)e-γx.

[5]

Model answer to Q 2(d): Bookwork Derivation

$$Js = \int_0^\infty Jc(x)\,dx$$

$$Js = \frac{Jc(0)}{-\gamma}\left[e^{-\gamma x}\right]_0^\infty$$

$$\therefore Jc(0) = \gamma Js$$

$$\gamma = \sigma_o Zs$$

[4]

Model answer to Q 2(e): Bookwork Derivation

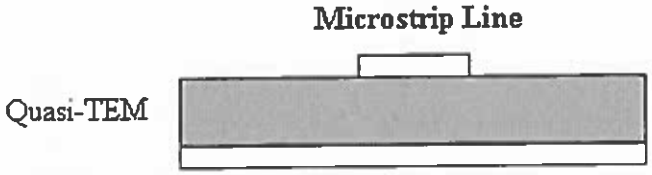
At the surface of the conductor, the conduction current leads the surface current by 45°, since Jc(0) = σo Zs Js = σo ($\sqrt{2}\, e^{+j\pi/4}$ Rs) Js

[4]

Answers

Model answer to Q 3(a): Bookwork

(i) microstrip line



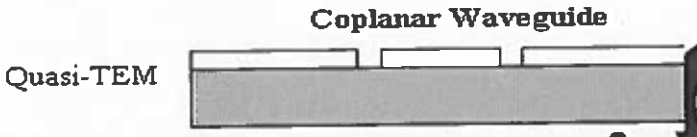
Advantages: easy to manufacture and many models available in CAD software

Disadvantages: frequency dispersive and requires through substrate vias

Applications: general interconnect and impedance matching

[2]

(ii) coplanar waveguide



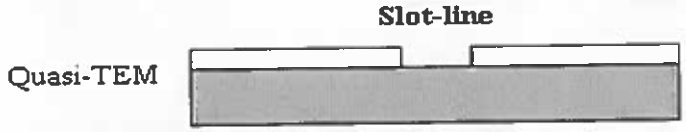
Advantages: easy to manufacture and no through substrate vias are required for grounding

Disadvantages: multi-moding is a problem at discontinuities and models are not so easy to find in CAD software

Applications: highly integrated circuits and microwave circuits

[2]

(iii) slotline



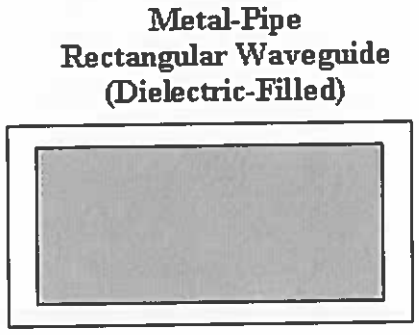
Advantages: easy to manufacture and easy to implement balanced signal lines

Disadvantages: multi-moding is a problem at discontinuities and models are not so easy to find in CAD software

Applications: feed lines for balanced antennas and balanced mixers

[2]

(iv) dielectric-filled metal-pipe rectangular waveguide



Advantages: very low loss and very high power handling

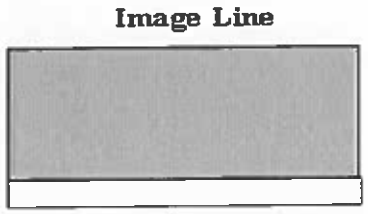
Answers

Disadvantages: frequency dispersive and difficult to integrate active devices

Applications: radio astronomy and radar

[2]

(v) image line



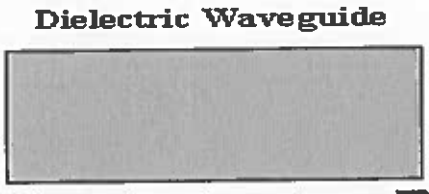
Advantages: very low loss and easy to manufacture

Disadvantages: frequency dispersive and difficult to integrate active devices

Applications: power couplers and low loss millimetre-wave interconnects

[2]

(vi) dielectric waveguide



Advantages: very low loss and easy to manufacture

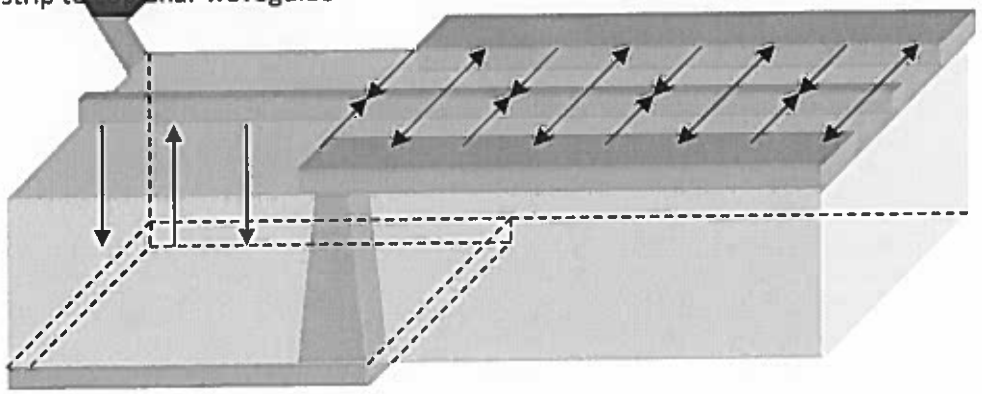
Disadvantages: poor isolation and difficult to integrate active devices

Applications: power couplers and low loss millimetre-wave interconnects

[2]

Model answers to Q 3(b) Bookwork

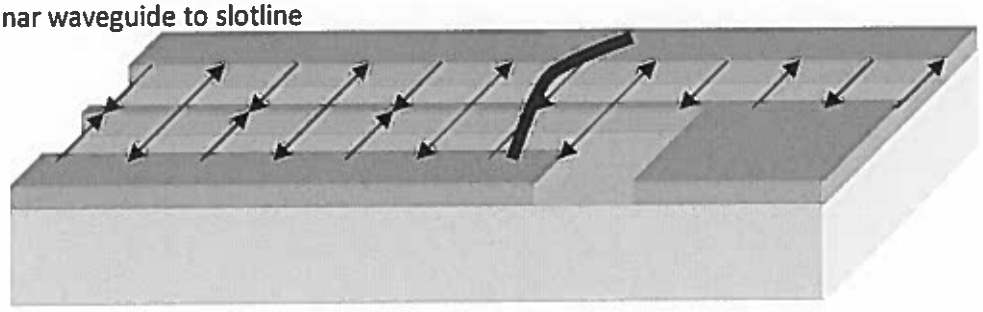
(i) microstrip to coplanar waveguide



[1]

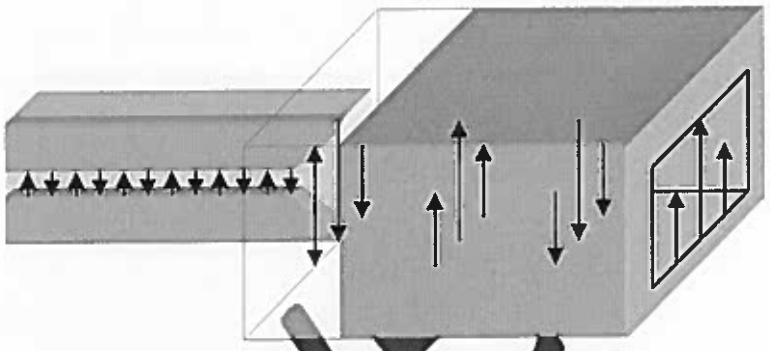
Answers

(ii) coplanar waveguide to slotline



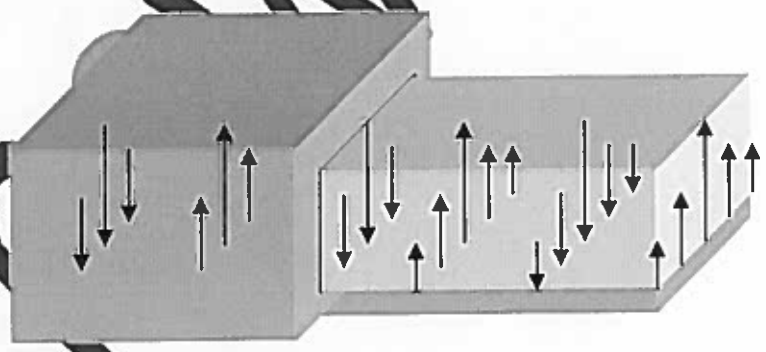
[1]

(iii) slotline to meta-pipe rectangular waveguide



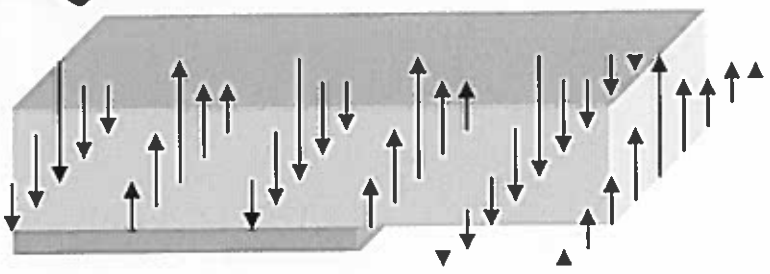
[1]

(iv) dielectric-filled metal-pipe rectangular waveguide to image line



[1]

(v) image line to dielectric waveguide



[1]

ARMC

Model answer to Q 1(c): Bookwork

At transitions between different types of guided-wave structures, or indeed at any physical and/ or electrical discontinuities, there may be the possibility of creating unwanted modes of propagation. If these unwanted modes are degenerate in nature then they must be suppressed (e.g. by using bond-

wires/underpasses or resonators). If the unwanted modes are evanescent in nature then they will die away within a few guided wavelengths, and so extra lengths of transmission lines are required to allow this to happen so that they do not have enough energy to excite degenerate modes.

[3]

Model answer to Q 4(a): Bookwork and Derivation Exercise

The voltage and current on the line can be represented as :

$$V(z) = V_+\left(e^{-\gamma z} + \rho(0)e^{+\gamma z}\right)$$
$$I(z) = I_+\left(e^{-\gamma z} - \rho(0)e^{+\gamma z}\right)$$

$$It\ can\ be\ found\ that: V_+ = 0.5\left(V(0) + ZoI(0)\right) \quad and \quad V_- = 0.5\left(V(0) - ZoI(0)\right)$$

$$\therefore incident\ wave\ power, P_+ = \frac{|V_+|^2}{Zo} \quad and \quad reflected\ wave\ power\ P_- = \frac{|V_-|^2}{Zo}$$

If Zo is taken to be purely real, the time-average power flow along the line is:

$$\mathrm{P(z)} = \mathrm{Re}\{V(z)I(z)^*\} = \mathrm{Re}\left\{V_+\left(e^{-\gamma z} + \rho(0)e^{+\gamma z}\right)I_+\left(e^{-\gamma z} - \rho(0)e^{+\gamma z}\right)^*\right\}$$

$$where, \rho(z) = \rho(0)e^{+2\gamma z} \equiv \rho(0)e^{+j2\beta z} \ for\ a\ lossless\ line$$

$$\mathrm{P(z)} = \mathrm{Re}\left\{\frac{|V_+|^2}{Zo}\left(1+\rho(z)\right)\left(1-\rho(z)\right)^*\right\} = \mathrm{Re}\left\{\frac{|V_+|^2}{Zo}\left(1+\rho(z)\right)\left(1-\rho(z)^*\right)\right\} = \frac{|V_+|^2}{Zo}\left(1-|\rho(z)|^2\right)$$

$$but, |\rho(z)| = |\rho(0)| \ for\ a\ lossless\ transmission\ line$$

$$\therefore \mathrm{P(z)} = \frac{|V_+|^2}{Zo}\left(1-|\rho(0)|^2\right) = P_+\left(1-\frac{P_-}{P_+}\right) = \frac{|V_+|^2}{Zo}\left(1-|\rho(0)|^2\right) = P_+\left(1-\frac{P_-}{P_+}\right) = \left(P_+ - P_-\right)$$

This shows that, for a lossless transmission line, time-average power flow is independent of the line length and is equal to the incident wave power minus the reflected wave power.

[5]

Model answer to Q 4(b): Bookwork and Derivation Exercise

The guided wavelength, λg, is defined as the distance between two successive points of equal phase on the wave at a fixed instance in time. The phase velocity of a wave is defined as the speed at which a constant phase point travels down the line. Frequency dispersion is said to occur when $\beta \neq \omega.constant$. Dispersion can occur when $vp = f(\omega)$, i.e. when $Dk = f(\omega)$. It can be shown that zero dispersion in a lossy line can also occur, but only when $RC = GL$:

$$\gamma^2 = \left(R + j\omega L\right)\left(G + j\omega C\right) \quad and \quad RC = GL$$

$$\therefore \alpha(\omega) = \alpha(0) = \sqrt{RG} \neq f(\omega) \quad and \quad \beta = \omega\sqrt{LC}$$

$$also, Group\ Velocity, Vg = \frac{\partial\omega}{\partial\beta} = \frac{1}{\sqrt{LC}} \equiv vp \neq f(\omega)$$

[5]

Answers

Model answer to Q 4(c): Bookwork

$$Zin = j\omega L + \frac{Zo\dfrac{1}{j\omega C}}{Zo + \dfrac{1}{j\omega C}} \equiv Zo$$

$$\therefore Zo = \frac{j\omega L}{2}\left(1 \mp \sqrt{1 - \frac{4}{\omega^2 LC}}\right) = \frac{j\omega L}{2}\left(1 \mp \sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2}\right)$$

$$\therefore \textit{Cut-off frequency, } fc = \frac{1}{\pi\sqrt{LC}} \text{ representing the bandwidth, i.e. when } \frac{4}{\omega^2 LC} \equiv 1$$

$$Zo = \begin{cases} \sqrt{\dfrac{L}{C}} & when \quad \omega << \omega_c \quad i.e.\ purely\ real \\ Complex & when \quad 0 << \omega < \omega_c \\ j\sqrt{\dfrac{L}{C}} & when \quad \omega = \omega_c \quad i.e.\ purely\ imaginary \\ j\omega L & when \quad \omega >> \omega_c \quad i.e.\ purely\ imaginary \end{cases}$$

Note that this cut-off frequency is also referred to as the Bragg frequency, which is twice the cut-off frequency of a traditional lossless LC filter.

[5]

Model answer to Q 4(d): Computed Example

$$Zin(\omega_c) = jZo(\omega << \omega_c) \quad and \quad |\rho(\omega_c)|^2 = 1 \Rightarrow 0\,dB \quad and \quad |\tau(\omega_c)|^2 = 1 - |\rho(\omega_c)|^2 = 0 \Rightarrow -\infty\,dB$$

[5]

Model answer to Q 5(a): Bookwork

The structure has the ideal (i.e. lossless) dominant-mode guided wavelength given by the following textbook expression:

$$\lambda_{g_ideal} = \frac{\lambda_o}{\sqrt{1 - \left(\dfrac{\lambda_o}{\lambda_c}\right)^2}} = \frac{\lambda_o}{\sqrt{1 - \left(\dfrac{f_c}{f}\right)^2}}$$

where, λ_o is the free space wavelength; $\lambda_c = 2a$ is the ideal cut-off wavelength; a is the internal width dimension of the RWG; $f_c = c/2a$ is the ideal cut-off frequency for the dominant TE_{10} mode; and c is the speed of light in free space.

[3]

Model answer to Q 5(b): Calculated Example

For a = 200 μm: f_c = c/(2a) =750 GHz and $\lambda_{g_ideal} = 566.7\,\mu m$

[2]

Model answer to Q 5(c): Bookwork

The corresponding textbook expression for the resonant frequencies for the TE_{mnl} modes in an ideal (i.e. lossless) cavity is given by:

Answ

$$\gamma = \hat{x}\gamma_x + \hat{y}\gamma_x + \hat{z}\gamma_z = jk$$

$$k = |\gamma| = \sqrt{k_x^{\,2} + k_y^{\,2} + k_z^{\,2}}$$

$$k_x = \frac{m2\pi}{\lambda_x} \quad \lambda_x = 2a; \quad k_y = \frac{n2\pi}{\lambda_y} \quad \lambda_x = 2b; \quad k_z = \frac{l2\pi}{\lambda_z} \quad \lambda_z = 2a;$$

$$k_{mnl_ideal} = \frac{2\pi f_{mnl_ideal}}{c} \quad \therefore f_{mnl_ideal} = \frac{c}{2\pi}\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{l\pi}{d}\right)^2}$$

[4]

Model answer to Q 5(d): New Derivation

For the dominant TE_{101} mode this becomes:

$$f_{101_ideal} = \frac{c}{2\pi}\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{l\pi}{d}\right)^2} = \sqrt{\frac{3}{8}}\left(\frac{c}{a}\right) \text{ when } d = \sqrt{2}a$$

$$f_{101_ideal} = \sqrt{1.5}f_c \text{ when } d = \sqrt{2}a$$

[2]

Model answer to Q 5(e): Computed Example

For a = 200 μm: b = 100 μm, d = [illegible] f_{101_ideal} = 918 GHz

[2]

Model answer to Q 5(f): Bookwork

$$Q_U = 2\pi \frac{Time - average\ energy\ stored\ at\ [illegible]\ frequency}{Energy\ dissipated\ in\ one\ [illegible]\ this}$$ (Dimensionless)

$$Q_{Uo} = \omega_{oo} \frac{[illegible]}{P_{DISS}}$$ [illegible]onless)

[1]

Model answer to Q 5(g): New Derivation

$$\lambda_{101_ideal} \equiv \frac{2\pi c}{[illegible]} \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + [illegible]^2 + \left(\frac{l}{d}\right)^2}} = 2\sqrt{\frac{2}{3}}a \quad for \quad d = \sqrt{2}a$$

[2]

Model answer to Q 5(h): Calculated Example

$$\delta = \sqrt{\frac{2}{\omega_{101_ideal}\mu_o\sigma_o}} = 78nm \quad at\ 918\ GHz$$

[2]

Model answer to Q 5(i): Calculated Example

$$Q_U = \frac{\lambda_{101_ideal}}{4\delta}\left\{\frac{2b\left(a^2+d^2\right)^{\frac{3}{2}}}{\left[2b\left(a^3+d^3\right)+ad\left(a^2+d^2\right)\right]}\right\} = \frac{2}{\delta_{So}}\left(\frac{3\sqrt{2}a}{4(5\sqrt{2}+1)}\right) \quad for \quad d = \sqrt{2}a$$

$$\therefore Q \approx 674 \quad for\ a = 200\mu m\ and\ d = \sqrt{2}a$$

[2]

Model answer to Q 6(a): Bookwork

In the kHz region, overly-priced Hi-Fi valve amplifiers are still commercially available.

In the MHz region, the cathode-ray tube found in domestic television receivers is still being used in the technologically backward western world.

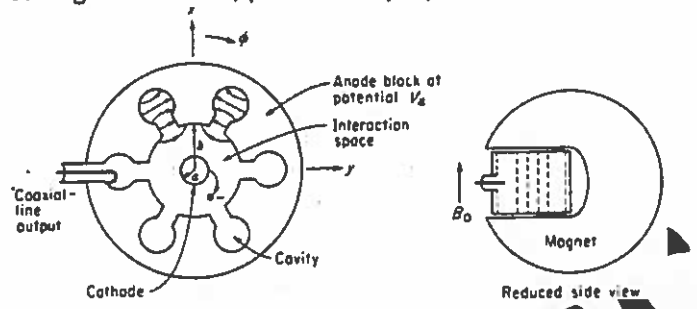
In the GHz part of the spectrum, the 2.54 GHz magnetron is found in all domestic microwave ovens.

Answers

[3]

Model answer to Q 6(b): Bookwork

- The cylindrical surface of the centre electrode is an electron-emitting cathode. The surrounding block forms the anode, containing N-resonant cavities and these are separated by N-segments. The axial length (i.e. cavity depth) is $\lambda o/4$. The complete structure is immersed in a strong axial H-field, produced by a permanent magnet.



- The magnetron is a self-excited oscillator that converts applied DC input power into pulsed RF output power. DC-to-RF conversion takes place in the interaction space between the cathode and the anode. The anode is constructed to form a microwave circuit consisting of a transmission line loaded with cavities. Initially a DC voltage is applied between the cathode and the microwave circuit. As the DC voltage is increased, electrons leave the heated cathode and interact with the H-field, causing them to circle the cathode. As the voltage is further increased, the angular velocity of the electrons becomes equal to that of the RF wave travelling along the microwave circuit. When this occurs, there is a spontaneous interaction between the electrons and the microwave circuit – resulting in spokes of space charge that induce microwave power into the slow-wave circuit. Here, electrons move through the RF-field of the resonant cavities. These electrons deliver energy to the resonators, which sustain the oscillations. The RF power is extracted by loose coupling into one of the cavities.

[10]

Model answer to Q 6(c): Bookwork

- The normal π-mode of operation (as shown above, right) can be regarded as being formed by the superposition of two waves rotating in opposite directions. For an *N*-cavity magnetron, the angular velocity at which both the electromagnetic wave and the electron spokes rotate is equal to $2\omega_0/N$.

- Modes other than the wanted π-mode are suppressed, by making an electrical connection to alternating segments, using two 'strapping rings' at both ends of the block. At higher frequencies, alternate cavities are made with differing depths – forming the 'rising sun' block. Both of these approaches have the effect of increasing the frequency separation between the wanted and the unwanted modes.

[5]

Model answer to Q 6(d): Bookwork

- Increasing the H-field requires an increased voltage for a given current, and results in a higher RF power output and efficiency.

- Pulsed applications: e.g. ~ 200 KW for airborne radar and ~ 20 MW for ground-based radar. CW applications: e.g. Diathermy and Domestic Microwave Ovens (<1 KW)

[2]

Answers